



POLICY

Recognising and Responding to Acute Deterioration

Scope (Staff):	All staff
Scope (Area):	PCH & Neonatology and Acute CAMHS (Out of Scope for CACH and CAMHS-CH)

Child Safe Organisation Statement of Commitment

CAHS commits to being a child safe organisation by applying the National Principles for Child Safe Organisations. This is a commitment to a strong culture supported by robust policies and procedures to reduce the likelihood of harm to children and young people.

This document should be read in conjunction with this [disclaimer](#) and the WA Health [Recognising and Responding to Acute Deterioration Policy](#)¹

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Aim

To outline the systems in place for staff to recognise and respond to acute deterioration in order to ensure a patient's deterioration is recognised promptly and appropriate action is taken.

Out of Scope

- **Child and Adolescent Community Health (CACH):** This policy does **NOT** apply to CACH staff and acute physiological deterioration in the community health setting.
 - For acute physiological deterioration in the community health setting, please refer to the CACH [Recognising and Responding to Acute Deterioration](#) policy.
- This policy does **not** apply to community CAMHS.

Risk

Failure to recognise and respond to acute deterioration may lead to adverse outcomes, increased length of stay and failure to rescue.

Definitions

Acute behavioural disturbance: conditions where a person behaves in a manner that may cause harm to themselves or others. It is characterised by a combination of delirium, agitation, aggressive behaviour, and autonomic dysfunction. Acute behavioural disturbance may have underlying physiological or psychiatric causes.

Acute deterioration: physiological, or mental state changes that may indicate a decline in the patient's health status. This may occur over a period of hours to days.

Clinical deterioration: includes physiological decompensation as well as acute changes in cognition and mental state.

Clinical observations: variables measured in the assessment for clinical deterioration; family concern, clinician concern, assessment of respiratory distress, respiratory rate, oxygen saturations, oxygen therapy, heart rate, blood pressure, central capillary refill time, pain score, level of consciousness/level of sedation, blood glucose level and temperature.

Delirium: a syndrome characterised by the acute onset of cerebral dysfunction with a change or fluctuation in baseline mental status, inattention and either disorganised thinking or an altered level of consciousness.²

Early warning systems: are tools used by hospital care teams to recognise early signs of clinical deterioration in order to initiate timely escalation of care.

Early warning score: the scoring system used for clinical observations routinely recorded at the patient's bedside. This scoring system triggers a predetermined response.

Escalation of care: is the recognition of deterioration and communication to a senior medical or nursing colleague, or rapid response team, resulting in a deterioration management plan.

Escalation pathway: refers to the process to be followed when completing clinical observations. The total early warning score determines the required clinical response which is outlined in the Early Warning Score Escalation Pathway and Sepsis Recognition Escalation Pathway.

Escalation system: refers to all the components used to recognise and respond to clinical deterioration and includes; the Paediatric Acute Recognition and Response Observation Tool (chart), the Early Warning Score Escalation Pathway, Sepsis Recognition Escalation Pathway, the PCH Sepsis Pathway, Aishwarya's CARE Call and structured communication tool iSoBAR.

Mental State deterioration: a negative change in mood or thinking, observed by a change in behaviour, perception, cognitive function, or emotional state.

Recognition: identification of clinical deterioration.

Response/Clinical response: actions taken to address identified clinical deterioration.

Sepsis: a dysregulated host response to infection leading to life threatening end organ dysfunction.

Septic Shock: sepsis with evidence of cardiovascular dysfunction. It is associated with high morbidity and mortality in children.

Sepsis Recognition Escalation Pathway: provides guidance on recognition of sepsis and appropriate escalation of care when sepsis is suspected. The sepsis pathway outlines criteria which trigger a sepsis review.

Sepsis review: consists of a senior clinician review to confirm or rule out a suspected sepsis diagnosis. The criteria which trigger a sepsis review are outlined in the Sepsis Recognition Escalation Pathway and the PCH Sepsis Pathway. A senior clinician at PCH is the Consultant or Fellow. If unavailable, or afterhours, this is the ED/STARS Registrar or most senior MET team member.

Track and trigger system: systems that rely on periodic measurement of clinical observations (tracking), with a predetermined response criterion (trigger) when a certain threshold is met.

Variable (weighted/unweighted): variables refer to the clinical observations recorded in the Paediatric Acute Recognition and Response Observation Tool. Weighted variables are those which are scored and contribute to the total early warning score. Unweighted variables are observations which are not scored but are still important indicators which need to be monitored and considered when assessing for clinical deterioration.

Principles

- CAHS will maintain systems for recognising and responding to acute deterioration in compliance with the following:
 - [National Safety and Quality Health Service Standard 8 Recognising and Responding to Acute Deterioration](#).²
 - Department of Health [Recognising and Responding to Acute Deterioration Policy](#).¹
 - [Australian Commission on Safety and Quality in Health Care Sepsis Clinical Care Standard](#).⁴
- Early recognition followed by prompt effective escalation and response can reduce adverse outcomes.

Monitoring for Acute Deterioration

- Monitoring and tracking the changes in vital signs and other clinical observations is a critical component of detecting acute deterioration.
 - Observation and response charts aid clinicians to observe trends in patient condition and elicit a timely response to deterioration.
- The Paediatric Acute Recognition and Response Observation Tool (PARROT) is the general observation chart approved for measuring and documenting clinical observations in inpatient and Emergency Department settings within CAHS. This tool forms part of the escalation system designed to recognise and respond to clinical deterioration.
 - Refer to the PCH [Observation, Monitoring and Paediatric Acute Recognition and Response Observation Tool \(PARROT\)](#).
- Other approved charts for monitoring and documenting physiological observations are used in the Paediatric Critical Care (PCC), Post Anaesthetic Care Unit (PACU), Operating Theatre (OT), and Neonates.
- The Agitation and Arousal Medication Chart is used to document patient arousal rating scales and assists clinicians with decisions regarding appropriate medication interventions when non-pharmacological de-escalation approaches are unsuccessful.
 - Refer to PCH [Pharmacy Arousal and Agitation Drug Management](#) guideline.

Escalation of Care

- The Escalation System includes all components required to recognise and respond to clinical deterioration.
 - Measurement and documentation of observations.
 - Communication framework using iSoBAR format.

- Refer to the CAHS [Communicating for Safety](#) policy.
- Family involvement (including history taking).
- Escalation pathways.
- Timely response to deterioration.
- Systems are in place to support escalation of care for the deteriorating patient in both inpatient and outpatient settings.
 - For escalation of care in PCH, refer to the following PCH policy documents:
 - [Resuscitation and Responding to Clinical Deterioration MET Review and Code Blue](#)
 - [Aishwarya's CARE Call: Family Escalation](#)
 - [STARS - Safety Team After-hours Response Service](#)
 - For escalation of care in CAMHS, refer to CAMHS [Physical Health Assessment and Monitoring](#) policy.
 - For escalation of care in Neonatology, refer to Neonatology [Recognising and Responding to Clinical Deterioration](#) guideline.
- The Paediatric Acute Recognition and Response Observation Tool includes two escalation pathways:
 - Early Warning Score Escalation Pathway.
 - Sepsis Recognition Escalation Pathway and the PCH Sepsis Pathway.
 - Refer to the PCH [Sepsis Recognition and Management](#) Guideline
- Other approved escalation pathways include:
 - [Major Trauma Management Pathway](#)
 - [Hospital in the Home \(HiTH\) Early Warning Score Escalation Pathway](#)
- Treatment-limiting care plans, known as [Paediatric Goals of Patient Care](#) exist to individualise escalation processes for patients with a life limiting condition.
 - Refer to the CAHS [Paediatric Goals of Patient Care](#) policy.
 - Refer to the PCH [End of Life Care](#) policy.

Monitoring for Mental State Deterioration and Delirium

- Determine baseline behaviour to assess level of deterioration from the patients usual presenting state.
- Assess for physiological/organic/iatrogenic cause.
- Consider Neurodevelopmental Disorders, baseline behaviours and effects of current contributing causes on mental state deterioration.

- Existing or emerging Mental Health diagnosis to be considered when assessing deterioration in current deterioration in Mental State.

Mental State Deterioration

- CAMHS staff must refer to the CAMHS [Risk Assessment and Management](#) policy, [Clinical Assessment](#) policy, and CAMHS Inpatient Unit [Progressive Risk Monitoring](#) work instruction for instruction regarding recognising and responding to acute mental health deterioration.
- For any indication of emerging delirium, staff are required to respond as per physiological or mental state acute deterioration processes.
- For acute behavioural disturbance, refer to
 - CAMHS [Arousal and Agitation Drug Management](#) policy
 - PCH [Acute Behavioural Disturbance Management of Patients with Developmental Disabilities](#) policy
 - CAMHS [Arousal and Agitation Management for Mental Health Inpatients](#) policy
- Refer to [Appendix 1](#) Recognising and responding to acute mental state deterioration flowchart for management strategies and escalation.
- A process exists for families / carers to escalate care via the Rapid Response System: Aishwarya's CARE Call.
 - Refer to the PCH [Aishwarya's CARE Call](#) policy.
 - For family/care escalation of care in Neonates located at Kind Edward Memorial Hospital (KEMH), refer to WNHS [Aishwarya's CARE Call](#).

Delirium Assessment and Management

- Delirium in children is an often under recognised but serious complication of hospitalisation⁵ often associated with underlying physical illness⁷.
- Delirium can occur in any setting, however neonatal and paediatric patients are deemed more at risk of emerging delirium during Post Anaesthetic Care Unit (PACU), neonatal intensive care and Paediatric Critical Care (PCC) episodes of care.
 - Emergence delirium is a transient state of marked irritation and disassociation after the discontinuation of anaesthesia in some patients. It is more likely to occur in paediatric patients between 2 and 5 years old following painful procedures using inhaled anaesthesia⁶.
 - Children in the PCC have an increased risk of delirium due to risk factors such as mechanical ventilation and medication requirements during their admission⁵.

- Children who present to the Emergency Department with acute behavioural changes must be assessed for delirium and the possible cause.
- Assessment for delirium must be documented on approved PCHN charts.
 - Delirium Assessment Tool CAPD (MR806.12)
 - Post Anaesthetic Care Unit Observation Chart (MR844.04)
 - Neonatal Abstinence Scoring System (MR495.00)
 - Progressive Mental Health Risk Assessment (MR508.00)
- All PCC and PACU patients must be monitored for delirium in accordance with the following:
 - Delirium Assessment and Management Guideline (Paediatric Critical Care Manual)
 - [PACU Discharge Criteria](#) (Perioperative Practice Manual)
- Other CAHS guidelines relevant for the assessment of altered mental status and delirium include:
 - [Comprehensive Care](#) (CAHS Policy Manual)
 - Comprehensive Admission and Screening Tool Forms prompt assessment of Behavioural, Emotional and Mental Health for 0-12 and 12 years and over.
 - [Behavioural Problems](#) (Emergency Department)
 - Alcohol and other drug (AOD) assessment and withdrawal management (Clinical Practice Manual)
 - PCH [Agitation and Arousal Drug Management](#) – Medication Management Manual
- The required level of observation will be determined by clinical judgement following assessment of risk of harm to self and others, clinical condition, cause and patient history.

Rapid Response Systems (RRS)

- A formal rapid response system is in place to facilitate rapid response for acute deterioration in inpatient settings.
 - The RRS at PCH comprises the Medical Emergency Team (MET) Review, Code Blue, and Safety Team After-hours Response Service (STARS).
 - For PCH, refer to the [Resuscitation and Responding to Clinical Deterioration. MET Review and Code Blue](#) procedure and the [STARS - Safety Team After-hours Response Service](#) guideline.
- The rapid response for acute deterioration in PCH outpatient settings is facilitated through activation of the Code Blue response.

- Critical care and specialist units such as Paediatric Critical Care (PCC), Post Anaesthetic Care Unit (PACU), Theatre, Neonatal Intensive Care Unit (NICU) and Emergency Department (ED) utilise their internal departmental RRS to address rapid or acute deterioration.
- For acute mental state deterioration response, refer to:
 - CAMHS [Crisis Connect Entry Protocol](#)
 - PCH [Psychiatric In-reach Team](#) Model of Care
- For other areas of PCHN refer to the following:
 - PCC [Code Blue – Responsibilities of PCC staff](#) protocol
 - Neonatal [Recognising and Responding to Clinical Deterioration](#) Clinical Practice Guideline.
 - PCH [Allergic Reactions and Anaphylaxis Management in Planned Allergy Challenges \(Immunology\)](#)

Clinical Communication

- The iSoBAR tool is the approved communication tool within Child and Adolescent Health Service (CAHS).
- Structured handover processes exist to transfer the responsibility and accountability for some or all aspects of care for a patient, or group of patients, on a temporary or permanent basis.
 - Refer to the CAHS [Communicating for Safety](#) policy.

Patients and Carers

- Patients and parents / carers must be informed about recognition and response systems, including how to raise concerns about their child's condition, and be involved in clinical handover.
 - Refer to the CAHS [Communicating for Safety](#) policy.
- Patients and parents / carers must also be made aware of and engaged in the following:
 - Information about possible deterioration should be sought from the patient and parent/carer when measuring clinical observations with the Paediatric Acute Recognition and Response Observation Tool using the family concern variable.
 - Information about deterioration should be communicated to the patient and parent / carer in a timely manner.
 - Communication and discussion should be undertaken with the patient, family or carer on any clinical decision making with regard to treatment planning, including where appropriate, a mechanism to communicate and

record their end of life wishes, limitations for escalation of treatment and the agreed levels of intervention; and

- Information provided on how to raise concerns about their child's condition in the hospital setting using [Aishwarya's CARE Call: Family Escalation](#):
 - A 3-step process which will initiate a review and assessment, and if required, escalation of care
 1. Talk to your child's nurse or doctor
 2. Talk to a senior staff member
 3. Make [Aishwarya's CARE Call: Family Escalation](#)
- Families can access Aishwarya's CARE Call information on the back of the admission packs, PES information system and other electronic safety screens. Flyers can be provided to families on inpatient areas and Emergency Department. *Tell us if you are worried* flyers and posters are also available for families.

Leadership and Governance

- CAHS and PCHN Committees exist to provide leadership and governance to the process and procedures for recognition and response to acute deterioration and these include:
 - PCHN Recognising and Responding to Acute Deterioration Committee;
 - PCH Resuscitation Committee; and
 - PCH Mortality and Morbidity Review Committee.

Education

- All staff are trained in, and must be compliant with, the systems in place to manage early recognition and response to acute deterioration within their scope of practice and employment. Refer to the following policy documents:
 - CAHS [Safety Skills Training](#) policy for further information.
 - [Resuscitation Training: Hospital Paediatric Life Support](#) procedure in the PCH Clinical Practice Manual.

Evaluation, Audit and Feedback

- Compliance with this policy is monitored and evaluated by the PCHN Recognising and Responding to Acute Deterioration Committee.
- CAHS audits, evaluates, and reports on the following performance measures:
 - Proportion of clinical staff that have completed training in:
 - Hospital Paediatric Life Support / Basic Life Support

- Escalation and PARROT
- PCH Sepsis Pathway
- Documentation of monitoring plans for inpatients.
- Completion rates of measurement and documentation of clinical observations.
- Number of and response to Aishwarya's CARE Calls, MET and Code Blue calls.
- Early Warning Scores and the escalation criteria will be evaluated in COBRA audits.
- Number of modifications to Early Warning Scores and if modifications are implemented in line with policy.
- Percentage of patients transferred to PCC post MET / CODE response.
- Number of inpatient cardiopulmonary arrests.
- Rates of unexpected and preventable mortality.
- Clinical documentation after rapid response system calls.
- Rate of patient and parent / carer awareness and utilisation of carer escalation process (Aishwarya's CARE Calls).
- PCH Sepsis Pathway Audit measures and reports against the [Sepsis Clinical Care Standard indicators](#).
- The PCHN Recognising and Responding to Acute Deterioration Committee reports to the PCHN Clinical Governance Committee.
- The Clinical Incident Management System (CIMS) is used to report, monitor, and investigate clinical incidences related to acute deterioration.
 - Refer to the CAHS [Clinical Incident Management](#) policy.
- The CAHS Safety, Quality and Performance Unit will provide monthly reports on reported incidents involving acute deterioration to the Recognising and Responding to Acute Deterioration Committee who will review and provide recommendations as and when required.
- The PCHN Recognising and Responding to Acute Deterioration Committee will identify trends in the above performance measures and recommend improvement strategies.

References and related external legislation, policies, and guidelines

1. [Recognising and Responding to Acute Deterioration Policy](#) (WA Health Mandatory Policy MP 0171/22)

2. National Safety and Quality Health Service Recognising and Responding to Acute Deterioration Standard (NSQHS Standards)
3. Australian Commission on Safety and Quality in Health Care National Consensus Statement: essential elements for recognising and responding to acute physiological deterioration 2nd Edition, January 2017
4. Australian Commission on Safety and Quality in Health Care Sepsis Clinical Care Standard
5. Delirium in Children: Identification, Prevention, and Management - PubMed (nih.gov)
6. Paediatric emergence delirium BJA Education Oxford Academic (oup.com)
7. A Clinical Pathway to Standardize Care of Children with Delirium in Paediatric Inpatient Settings. Silver G, Kearney JA, Bora S et al. Hospital Paediatrics Vol 9; Iss 11; Nov 2019.

Related CAHS internal policies, procedures and guidelines
Acute Behavioural Disturbance Management of Patients with Developmental Disabilities (PCH Clinical Practice Manual)
Aishwarya's CARE Call (WNHS Procedure)
Aishwarya's CARE Call: Family Escalation (PCH Operational Manual)
Arousal and Agitation Management for Mental Health Inpatients (CAMHS Policy Manual)
Behaviour Problems (PCH ED Guideline)
Clinical Assessment (CAMHS Policy Manual)
Calling for Assistance After Hours - PACU (PCH Perioperative Practice Manual)
Clinical Handover (CAMHS Policy Manual)
Clinical Incident Management (CAHS Policy Manual)
Code Blue – Responsibilities of PCC staff (PCH Paediatric Critical Care Manual)
Communicating for Safety (CAHS Policy Manual)
Comprehensive Care – CAHS Policy Manual
Crisis Connect Entry Protocol (PCH CAMHS Policy Manual)

Delirium Assessment and Management (PCH Paediatric Critical Care Manual)
End of Life Care (PCH Clinical Practice Manual)
First Aid (CAHS Policy Manual)
Observation, Monitoring and Paediatric Acute Recognition and Response Observation Tool (PARROT) (PCH Clinical Practice Manual)
PACU Discharge Criteria (Perioperative Practice Manual)
PCH Psychiatric In-Reach Team (CAMHS) Model of Care
Physical Health Assessment and Monitoring (CAMHS Policy Manual)
Recognising and Responding to Acute Deterioration (CACH Operational Policy Manual)
Recognising and Responding to Clinical Deterioration (Neonatal Clinical Practice Guideline)
Resuscitation and Responding to Clinical Deterioration. MET Review and Code Blue (PCH Clinical Practice Manual)
Resuscitation Training: Hospital Paediatric Life Support (PCH Clinical Practice Manual)
Risk Assessment and Management policy (CAMHS Policy Manual)
Safety Skills Training (CAHS Policy Manual)
Sepsis Recognition and Management Guideline (ED Guidelines)
STARS - Safety Team After-hours Response Service (PCH Operational Manual)

Useful resources (including related forms)
Australian Commission on Safety and Quality in Healthcare - Recognising and Responding to Clinical Deterioration
Delirium Assessment Tool CAPD (MR806.12) – Clinical Form
Escalation Mapping Template – Australian Commission on Safety and Quality in Healthcare
MET Review and Code Blue at PCH Take 5
Neonatal Abstinence Scoring System (MR495.00) – Clinical Form

NSQHS User Guide for Acute and Community Health Service Organisations that Provide Care for Children
Post Anaesthetic Care Unit Observation Chart (MR844.04) – Clinical Form
Progressive risk monitoring (CAMHS Inpatient Unit Work Instruction)
Rapid Fluid Resuscitation Take 5
Sepsis Take 5
Think Sepsis PCH Health Point Page
Progressive Mental Health Risk Assessment (MR508.00) – Clinical Forms

This document can be made available in alternative formats on request.

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Appendix 1: CAHS Mental State Deterioration and Escalation of Care Flowchart

